

Problem 2

What should be the ratio of distilled water to alcohol to obtain a liquid with a density of $\rho = 800 \text{ kg/m}^3$? When mixing water and alcohol, the total volume decreases by 3% compared to the initial sum of the component volumes.

Solution

Let: - $\rho_w = 1000 \text{ kg/m}^3$ be the density of water. - $\rho_a = 789 \text{ kg/m}^3$ be the density of alcohol. - V_w and V_a be the initial volumes of water and alcohol, respectively.

The masses of water and alcohol are:

$$m_w = \rho_w V_w, \quad m_a = \rho_a V_a.$$

The total mass is:

$$m_{\text{total}} = m_w + m_a = \rho_w V_w + \rho_a V_a.$$

Due to volume contraction, the final volume is:

$$V_{\text{final}} = 0.97(V_w + V_a).$$

The mixture density is given by:

$$\rho = \frac{m_{\text{total}}}{V_{\text{final}}}.$$

Substitute the expressions:

$$800 = \frac{1000V_w + 789V_a}{0.97(V_w + V_a)}.$$

Cross-multiply:

$$800 \times 0.97(V_w + V_a) = 1000V_w + 789V_a.$$

Expand:

$$776(V_w + V_a) = 1000V_w + 789V_a.$$

Distribute:

$$776V_w + 776V_a = 1000V_w + 789V_a.$$

Group like terms:

$$776V_w - 1000V_w = 789V_a - 776V_a.$$

Simplify:

$$-224V_w = 13V_a.$$

Solve for the ratio:

$$\frac{V_w}{V_a} = \frac{13}{224}.$$

Thus, the ratio of distilled water to alcohol should be:

$$V_w : V_a = 13 : 224.$$